Verifier Construction

Moving the Target into the Fast Loop

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Abstract

A companion paper defines the feedback horizon gap N_{proxy}: how many proxy-guided optimization updates occur before independent target-level evidence arrives (Zarncke 2026). That analysis treats the feedback architecture as given. This paper asks how the architecture can be changed.

Many domains now treated as objectively verifiable became so only after instruments, tests, endpoints, and institutions made previously latent errors observable. Call this *verifier construction*: building checkers, tests, measurements, experiments, surrogates, monitors, or independent audits that move target-relevant information into a shorter loop. Construction can reduce the horizon by earlier observation, cheaper repetition, or decomposition into shorter checkable pieces. Replacing a slow target with a fast surrogate is construction too. It shortens the decision loop while creating residual error that later optimization can exploit.

Which construction is available depends on the kind of feedback a domain can support. We distinguish six regimes, from exact symbolic checks to sparse and endogenous evaluation, and map sixteen AI and human domains onto that geometry. Human learning results caution that faster feedback is not automatically more informative. The AI–human comparison is a comparison of feedback architectures, not a claim that machine learning is accelerated human science.

The test of an evaluation suite is whether target-relevant errors become visible at roughly the rate at which optimization can discover and amplify them.

1 Introduction

The feedback horizon gap counts how many proxy-guided updates occur before independent target evidence can correct the process (Zarncke 2026). Large N_{proxy} is cheap when the fast evaluator is the target. It is costly when the fast evaluator is an imperfect proxy whose errors stay hidden.

That diagnosis leaves the architecture fixed. An optimizer receives a proxy every τ_{P}. More independent target evidence arrives after τ_{T}. If candidate generation and proxy evaluation get cheaper while τ_{T} stays put, N_{proxy} grows.

This paper treats the architecture as the intervention. Difficult problems often become tractable not only by searching faster but by making mistakes observable sooner and more faithfully. Formal proof checkers, unit tests, in-situ laboratory measurements, biomarkers, precursor analysis, and independent audits are instances of the same move: constructing an evidence channel.

The question is then not only

How large is N_{proxy}?

but

What would have to be built so that target-relevant error appears inside the fast loop?

Three distinctions matter immediately. First, a surrogate can shorten latency while losing target information. That is still construction, and it is the dangerous kind. Second, a verifier answers a claim. Lean checks formal validity, not faithful formalization. A unit test checks tested behavior, not security. Third, a verifier validated on randomly sampled candidates need not remain reliable on candidates selected to maximize it.

Section 2 restates N_{proxy} only as far as this argument needs. Section 3 develops construction. Section 4 classifies six feedback regimes. Section 5 maps sixteen domains. Section 6 notes that human learning tracks information, not mere speed. Section 7 records where the AI–human analogy is asymmetric.

2 The Horizon Being Closed

The companion paper defines the gap and the associated accumulation identity (Zarncke 2026). This section restates only what the present argument uses.

Let τ_{P} be the latency of the feedback actually used for optimization, and τ_{T} the latency until informative target-level evidence is available. Let N_{proxy} be the number of proxy-guided updates before the next independent target check. When updates are approximately periodic,

$$N_{\text{proxy}} \approx \left\lfloor \frac{\tau_{\text{T}}}{\tau_{\text{P}}} \right\rfloor .$$

Count updates directly when parallelism, batching, or variable step sizes make wall-clock time a poor proxy.

Write $L_{\text{T}}(\theta)$ for target loss and $\Delta_i = L_{\text{T}}(\theta_{i+1}) - L_{\text{T}}(\theta_i)$ for the target effect of proxy-guided update i. Then exactly

$$L_{\text{T}}(\theta_{N_{\text{proxy}}}) - L_{\text{T}}(\theta_0) = \sum_{i=0}^{N_{\text{proxy}}-1} \Delta_i .$$

The identity separates how long error can accumulate from whether it does. A perfect evaluator can make large N_{proxy} cheap. An exploitable proxy can make moderate N_{proxy} costly. Systematic per-update drift accumulates in N_{proxy}; persistent correlation among proxy errors makes the sum more dangerous than independent noise.

Two further latencies appear in the companion analysis: candidate-generation latency τ_{G}, and the latency τ_{C} until target evidence can materially correct the optimizer or system. Usually $\tau_{\text{G}} \le \tau_{\text{P}} \le \tau_{\text{T}} \le \tau_{\text{C}}$, but not always.

The rest of this paper asks how to change those latencies, especially τ_{T}, rather than how to interpret a fixed N_{proxy}.

3 Verifier Construction

Treating τ_{P} and τ_{T} as fixed hides the main historical pattern: the feedback architecture can be changed.

A useful intervention is therefore not only

generate better candidates,

but

$\boxed{\text{make mistakes observable sooner and more faithfully.}}$

We call this *verifier construction*. The term includes formal verifiers, tests, instruments, experiments, surrogate measurements, monitoring systems, and institutions that move target-relevant information into a shorter feedback loop.

3.1 Verification is constructed, not merely discovered

Many domains now treated as "objectively verifiable" became so only after substantial technical and institutional work. A physical quantity requires an operational measurement procedure. An experimental result requires controlled comparisons, calibration, and a way to distinguish signal from noise. A software property requires a specification and a test or proof obligation. A clinical claim requires an endpoint and a study design.

Thus the apparent binary distinction

$$\text{verifiable} \quad \text{vs.} \quad \text{unverifiable}$$

is often better represented as

$$\text{weakly observable} \rightarrow \text{instrumented} \rightarrow \text{repeatably measurable} \rightarrow \text{cheaply testable.}$$

Scientific instruments are especially clear examples. An instrument can turn a latent state of the world into a repeatedly accessible signal. The important improvement is not necessarily that the world changes faster. Instead,

$$I(F;T)$$

increases, or informative F becomes available at shorter latency.

The same transformation occurs in engineering when an expensive full-system failure is replaced by component tests, simulation, telemetry, or formal invariants.

3.2 Verifier construction can reduce the feedback horizon in three ways

The framework separates three distinct improvements.

Earlier observation. A new measurement can reveal a relevant effect before the final outcome occurs. Continuous telemetry, biomarkers, in-situ spectroscopy, and precursor events are examples. If the earlier measurement is genuinely informative about the target, this reduces effective τ_{T}.

Cheaper repetition. Automation can make the same target-relevant experiment repeatable at much higher throughput. A-Lab integrates planning, synthesis, and characterization and executed 353 experiments in 17 days (Szymanski et al. 2023). AlphaFlow similarly couples experiment selection with in-situ measurement (Volk et al. 2023). Neither system removes chemical or physical latency, but both remove substantial human handoff latency and increase the number of target-relevant observations per unit time.

Decomposition. A long-horizon target can sometimes be split into shorter independently checkable conditions. Software tests provide a familiar example. A requirement such as "this service remains correct in deployment" is not directly verifiable in one short experiment, but type checks, unit tests, property tests, integration tests, security tests, and monitoring each constrain different failure modes.

In the ideal case,

$$T \approx g(T_1, \ldots, T_m),$$

where each T_i has a much shorter verification horizon than T. The value of the decomposition depends on whether satisfying the pieces is sufficient for the whole. Missing interactions can recreate the original long-horizon problem.

3.3 Surrogates are compressed verifiers

A common way to shorten the feedback horizon is to replace a slow target T with an earlier observable P:

$$T \longrightarrow P \longrightarrow \text{fast optimization.}$$

This is verifier construction too, but it creates the horizon gap. A surrogate trades latency for approximation error.

Clinical medicine makes the trade explicit. Accelerated approval can rely on surrogate endpoints that are expected to predict clinical benefit, allowing decisions before survival or quality-of-life evidence fully matures (Gyawali, Hey, and Kesselheim 2019; Naci, Smalley, and Kesselheim 2017). Later studies sometimes fail to show corresponding patient-centered benefits (Liu, Kesselheim, and Cliff 2024). The surrogate has successfully shortened the decision loop while imperfectly shortening the *target* loop.

Reward models are the analogous construction in preference optimization. Expensive human judgments are compressed into a model that can be queried at machine speed. This creates enormous evaluation throughput but also creates an exploitable residual: continued optimization can increase proxy reward while independent evaluation degrades (Gao, Schulman, and Hilton 2022; Rafailov et al. 2024).

Verifier construction therefore creates a trade:

$$\text{latency reduction} \quad \text{versus} \quad \text{loss of target information.}$$

3.4 A verifier is useful only relative to a claim

There is no universal verifier for an artifact.

Lean verifies formal validity, not that the theorem is the one the human intended. A unit test verifies tested behavior, not global program correctness. A vulnerability reproduction harness verifies the presence of one exploit, not the absence of others. A biomarker measures a biological response, not necessarily patient benefit.

We can express this as a chain:

$$\boxed{\text{target specification} \to \text{observable} \to \text{verifier} \to \text{optimization.}}$$

Failure can occur at each arrow.

This also separates *certification* from *construction*. Given a candidate x, an inexpensive verifier may answer whether x satisfies a criterion without providing an efficient method for finding such an x. Formal theorem proving makes the distinction especially clear: proof checking is cheap compared with proof search. Verifier construction improves the selection landscape; it does not by itself solve the search problem.

3.5 Optimization changes the requirements on the verifier

A verifier that works for randomly sampled candidates need not remain reliable on candidates selected specifically to maximize it. Let $P_0(x)$ be the distribution on which an evaluator was validated and $P_k(x)$ the distribution after k rounds of optimization. What matters is not only

$$\mathrm{Corr}_{P_0}(F, T),$$

but whether the relationship survives the shift

$$P_0 \to P_1 \to \cdots \to P_K.$$

Reward-model overoptimization provides direct evidence of this problem (Gao, Schulman, and Hilton 2022). Continued policy optimization moves samples into regions where the learned evaluator is less reliable. More generally, optimization searches for the residual error of the verifier.

A useful verifier for a powerful optimizer must therefore satisfy a stronger criterion than ordinary predictive accuracy:

high fidelity under the distribution induced by optimizing the verifier.

This requirement becomes especially difficult when the optimizer can influence the measurement process itself.

3.6 Independent correction is part of verifier design

Verifier construction should therefore not aim only at making a single evaluator faster. It can also construct *independent* correction channels.

Suppose a fast evaluator F_1 guides most optimization, while evaluators $F_2, \ldots, F_m$ are queried less frequently. The secondary evaluators are valuable when their errors are not strongly correlated:

$$\mathrm{Cov}(\epsilon_i, \epsilon_j) \text{ is small}, \qquad \epsilon_i = F_i - T.$$

A second evaluator produced from the same data, architecture, and objective may add little independent information even if its outputs look different. Independence is therefore a causal property of the evaluation process, not merely an ensemble size.

Human institutions often approximate this architecture through replication, independent audits, adversarial review, appeals, and multiple measurement methods. These mechanisms are imperfect, but their structural purpose is relevant here: they prevent the process being optimized from relying indefinitely on one unchecked error channel.

3.7 Verifier construction is the main escape from a growing horizon gap

If AI reduces τ_{G} and τ_{P} while τ_{T} remains fixed, then N_{proxy} grows. One response is to slow optimization. A more powerful response, when possible, is to redesign the evidence channel:

$$\begin{gathered} \tau_{\mathrm{G}} \downarrow \qquad \tau_{\mathrm{P}} \downarrow \qquad \tau_{\mathrm{T}} \text{ fixed} \\ \Downarrow \\ N_{\mathrm{proxy}} \uparrow \end{gathered}$$

versus

$$\begin{gathered} \tau_{\mathrm{G}} \downarrow \qquad\qquad \tau_{\mathrm{P}} \downarrow \qquad\qquad \tau_{\mathrm{T}} \downarrow \\ \Downarrow \\ N_{\mathrm{proxy}} \text{ controlled.} \end{gathered}$$

This gives a different interpretation of scientific and engineering progress. Capability improvements search the solution space faster. Measurement, instrumentation, testing, and experimental design make the consequences of that search legible faster.

The two are complements. Fast search with a strong verifier produces the favorable regime represented by theorem proving and well-tested software. Fast search with a weak but rapidly computed proxy produces the regime represented by reward-model overoptimization.

For AI safety, the practical question is therefore not merely whether an evaluation suite exists. It is:

> Does the evaluation architecture make target-relevant errors visible at roughly the rate at which optimization can discover and amplify them?

If not, increasing evaluator throughput alone may make the proxy loop faster without making the correction loop faster. The feedback horizon has not actually been closed.

4 Feedback Regimes

The distinction between proxy and target feedback becomes more useful when combined with the *kind* of feedback available. A short feedback horizon is not equally valuable in every regime, and the available construction is not the same in every regime. What matters is both how rapidly evidence arrives and what that evidence establishes.

We distinguish six broad regimes. They are not mutually exclusive. A single system can move through several of them as an artifact progresses from simulation to deployment.

Exact symbolic feedback. A candidate is checked against a formally specified criterion, and acceptance or rejection follows mechanically. Formal theorem proving is the clearest example. Once a mathematical statement has been faithfully represented in Lean, a proof checker provides an unusually tight relationship between the feedback used during search and the local target of formal validity (Hubert et al. 2025). Rule-complete games have a similar structure at the level of terminal outcomes: the rules determine whether a position is legal and who won (Silver, Schrittwieser, Simonyan, et al. 2017).

This regime can support extremely large search because verification is cheap relative to generation. Its main residual failure lies outside the verifier: the formal statement, game objective, or specification may fail to represent what was intended. Thus exact verification solves

$$\text{“Does } x \text{ satisfy specification } S\text{?”}$$

without solving

$$\text{“Is } S \text{ the right specification?”.}$$

Executable feedback. The candidate is run, and behavior is compared with tests, assertions, constraints, or measured performance. Compilation, unit tests, property tests, exploit reproduction, and hardware benchmarks belong here. Execution feedback enables LLMs to debug generated programs iteratively (Chen et al. 2023). Algorithm optimization can similarly combine cheap correctness checks with less frequent direct measurements of runtime (Mankowitz, Michi, Zitnick, et al. 2023).

Executable feedback differs from exact symbolic feedback because the tested behavior is normally only a sample of possible behavior. A program can pass all available tests while remaining insecure, brittle, or incorrect outside the tested distribution. The feedback horizon is therefore property-specific:

$$\tau_{\mathrm{T}}^{\text{tested property}} \ll \tau_{\mathrm{T}}^{\text{unobserved property}}.$$

There is no single feedback horizon for “the program.”

Repeated empirical feedback. The system acts on the world and obtains direct measurements sufficiently quickly to guide the next action. Closed-loop chemistry and autonomous materials laboratories are increasingly close to this regime. AlphaFlow uses in-situ measurements to guide subsequent experiments (Volk et al. 2023), while A-Lab integrates experiment selection, robotic execution, synthesis, and characterization in one autonomous loop (Szymanski et al. 2023).

Here the verifier is not purely computational. Nature performs part of the computation. The cycle therefore has a physical floor set by manipulation, reaction, growth, fabrication, measurement, or other dynamics of the system being studied. Automation can remove human handoffs, but it cannot in general make $\tau_{\mathrm{T}} \to 0$.

Noisy statistical feedback. Many important outcomes are not determined by one observation. Evidence accumulates over repeated measurements, noisy labels, heterogeneous cases, or stochastic outcomes. Preference learning and clinical trials are examples. A reward model compresses many human judgments into a rapidly queried statistical evaluator; clinical surrogate endpoints compress later patient outcomes into earlier measurements.

In this regime, a feedback event need not be wrong in isolation. The problem is that its evidential relationship to the target is probabilistic:

$$P(T \mid F) \neq 0, 1.$$

Repeated optimization can therefore exploit stable residual error in the mapping from F to T. Reward-model overoptimization demonstrates this directly: independently evaluated quality can fall while the learned reward continues to rise (Gao, Schulman, and Hilton 2022; Rafailov et al. 2024). Clinical surrogate endpoints show the same structural distinction outside AI: an earlier endpoint can be useful while remaining an imperfect predictor of survival or quality of life (Gyawali, Hey, and Kesselheim 2019; Naci, Smalley, and Kesselheim 2017; Liu, Kesselheim, and Cliff 2024).

Sparse historical feedback. Some targets are observed only after long delays or rare events. Examples include major accidents, long-horizon institutional outcomes, macroeconomic policy effects, and sufficiently rare safety failures. In such settings it may be impossible to generate repeated target-level observations on demand.

The main difficulty is therefore not merely noisy evaluation but low target-observation rate. If catastrophic failures occur once per 10^6 system-hours, a million hours of safe operation is still only weak evidence about failure probabilities much below that scale. Mature safety fields respond by constructing intermediate evidence: near misses, precursor events, stress tests, component reliability estimates, simulation, and probabilistic risk models. These do not eliminate the long target horizon. They attempt to infer the target from more frequent observations.

Endogenous or reflexive feedback. Finally, the system being optimized may influence the process that evaluates it. Recommendation algorithms affect the users whose clicks constitute future training data. Economic policy changes the economy from which subsequent measurements are taken. A strategically capable AI may alter, anticipate, select for, or otherwise condition on its evaluators.

The distinction is structural. In the earlier regimes we can approximately write

$$\theta \longrightarrow \text{outcome} \longrightarrow F,$$

with F treated as an external observation. In a reflexive regime,

$$\theta \longrightarrow \{\text{outcome, measurement process, future data}\} \longrightarrow F.$$

The evaluator is now part of the environment being optimized.

This is more demanding than ordinary proxy error. Even an initially high-fidelity evaluator can become less informative under optimization because the distribution or the measurement process changes. Real-time engagement optimization provides a mild example: clicks can be measured almost instantly, yet optimizing them need not optimize the user's ideal preferences or welfare (Khambatta et al. 2023). More adversarial forms are a central concern in AI alignment, but are not yet supported by a comparably clean empirical literature.

The feedback-horizon quantity N_{proxy} is most informative when read together with this classification. Large N_{proxy} under exact symbolic feedback can be benign. Moderate N_{proxy} under a systematically exploitable statistical evaluator can already be problematic. Under sparse or endogenous feedback, even defining an independent target check may be the hard part.

The same classification constrains construction. Exact symbolic feedback is already constructed. Sparse and endogenous regimes may require inventing an intermediate evidence channel before N_{proxy} is even well defined.

5 Matched Feedback Horizons

Table 1 compares fast proxy loops, current AI cycles, human or classical cycles, target-evidence latency τ_{T}, and approximate N_{proxy} across sixteen domains. Bracket tags mark provenance: [M] directly measured or reported in the cited source; [D] simple derivation from reported quantities; [E] order-of-magnitude or workflow estimate; [NQ] not defensibly quantified from available literature; [S] structural analogy rather than direct empirical evidence.

The table is a map of where construction has already placed the target in the fast loop, where it has produced a surrogate, and where an independent target check is still poorly defined. It is not a claim that every row obeys the same quantitative law.

Table 1: Matched feedback structures across AI and human systems. τ_{G} is candidate-generation latency, τ_{P} the latency of feedback actually used for selection or updating, and τ_{T} the latency until sufficiently independent evidence about the target is available. N_{proxy} is reported only where the notion of repeated proxy-guided updates before a target check is meaningful. Platform throughput is not silently converted into serial latency. Domain-by-domain readings appear in Subsection 5.1.

Domain	**Fast loop / proxy**	**Current AI cycle**	**Human / classical cycle**	**Target-level evidence τ_{T}**	**Limiting cycle**	**Approx. N_{proxy} / status**
Formal theorem proving	Formal proof checker, e.g. Lean acceptance/rejection.	Lean verification is inside the search loop. AlphaProof used millions of formally checked problem variants; difficult IMO solutions required 2–3 days of test-time RL end-to-end [M] (Hubert et al. 2025).	Human proof construction and review are typically hours to days for nontrivial problems [E]; no matched cycle-time study is available.	For the target "valid proof of this formal statement", target evidence arrives when the proof checker accepts or rejects.	Proof-checking computation.	Potentially enormous N_{proxy} of search steps, but effectively $N_{\mathrm{proxy}} \simeq 1$ relative to formal validity.
Executable software	Compilation, execution, unit tests, property tests.	LLMs can repeatedly execute and self-debug generated programs; per-iteration latency is implementation-dependent [NQ] (Chen et al. 2023).	In a randomized coding experiment, the control group required 2 h 41 min on average versus 1 h 11 min with Copilot [M] (Peng et al. 2023).	Seconds or minutes for covered functional properties [E], but security, maintainability, performance under real workload, and rare bugs can remain latent much longer.	Execution plus test runtime for covered properties.	Property-dependent.
Rule-complete games	Exact game rules and terminal win/loss.	AlphaGo Zero learned entirely from self-play; every completed game provides an exact terminal result [M] (Silver, Schrittwieser, Simonyan, et al. 2017).	Humans normally receive a correct game result after each played game; wall-clock duration depends on the game and time control [NQ].	The task-level target, winning under the stated rules, is observed at game termination.	Simulator/game execution.	Approximately one terminal target check per completed game; many internal search steps may occur between them.

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Table 1 continued

Domain	**Fast loop / proxy**	**Current AI cycle**	**Human / classical cycle**	**Target-level evidence τ_T**	**Limiting cycle**	**Approx. N_{proxy} / status**
Low-level algorithm optimization	Correctness tests plus predicted or measured runtime.	AlphaDev trained for at most about 2 days in the reported tasks [M], explored at most 12 million programs in its worst reported case [M], and directly measured actual latency for less than 0.002% of generated programs [M] (Mankowitz, Michi, Zitnick, et al. 2023).	The optimized libc++ subroutines had not been changed for over a decade before the reported replacement [M] (Mankowitz, Michi, Zitnick, et al. 2023).	Correctness can be tested cheaply; true hardware latency is more expensive and therefore sampled sparsely.	Program execution on target hardware.	Very large between direct latency measurements; exact N_{proxy} depends on the training trajectory.
Chip floorplanning	Estimated wire-length, congestion, power/performance/area surrogates.	RL-generated floorplans in under 6 hours [M] (Mirhoseini, Goldie, Yazgan, et al. 2021).	The source describes conventional floorplanning as requiring months of intense effort by physical-design engineers [M] (Mirhoseini, Goldie, Yazgan, et al. 2021).	Full downstream place-and-route, timing closure, physical verification, fabrication, and silicon performance provide progressively stronger checks.	EDA computation initially; ultimately fabrication and physical measurement.	[NQ]
Weather forecasting	Forecast generation and retrospective scoring against known historical weather.	GraphCast generates a 10-day global forecast in under 60 seconds [M] (Lam et al. 2023).	The reported HRES numerical forecast requires hours on a supercomputer for a comparable 10-day prediction [M] (Lam et al. 2023).	A 6-hour forecast can only be prospectively verified after 6 hours; a 10-day forecast after 10 days.	The physical evolution of the atmosphere.	Not an optimization N_{proxy} for a fixed forecaster; the generation/target ratio can nevertheless exceed 10^4 for a 10-day forecast.
General forecasting	Brier/log scores once questions resolve; before resolution, forecasters can update probabilities from new information.	ForecastBench starts new rounds every two weeks and samples 1,000 questions for LLMs and 200 for humans [M] (Karger et al. 2025).	Expert human forecasters are evaluated on the same future questions; the original study found experts outperforming the top tested LLM [M] (Karger et al. 2025).	Questions are explicitly about future events whose answers are unknown at submission; resolution is therefore event-limited.	Occurrence and reliable resolution of the event.	[NQ]; varies by question horizon.

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Table 1 continued

Domain	**Fast loop / proxy**	**Current AI cycle**	**Human / classical cycle**	**Target-level evidence τ_T**	**Limiting cycle**	**Approx. N_{proxy} / status**
Autonomous materials synthesis	Synthesis outcome and automated characterization guide the next experiment.	A-Lab executed 353 experiments over 17 days [M], about 20.8 experiments/day platform throughput [D] (Szymanski et al. 2023).	The paper motivates automation partly by the slower conventional human-mediated workflow, but does not provide a clean matched per-experiment human cycle [NQ].	X-ray diffraction and other characterization can close the laboratory loop after synthesis; broader material performance may require additional testing.	Robotic handling, heating/reaction time, cooling and characterization.	Approximately one measured laboratory check per experiment for the measured property; do not infer serial latency from throughput.
Closed-loop chemistry	In-situ spectroscopy and reaction measurements feed subsequent experiment selection.	AlphaFlow used roughly 700 injection steps per campaign and produced more than 9,000 experimental conditions for learning [M] (Volk et al. 2023).	Conventional workflows require human transfer between design, execution, characterization and next-step selection; a matched wall-clock comparison is not reported [NQ].	The measured chemical state is available after the relevant reaction and characterization step.	Chemical kinetics, fluid handling and measurement.	Near one for the measured in-situ property; larger for unmeasured downstream properties.
Drug development and clinical medicine	Biomarkers, tumor response, progression-free survival and other surrogate endpoints can guide decisions earlier than patient-centered outcomes.	AI may accelerate candidate generation, but no general AI τ_P is defensibly comparable across drug programs [NQ].	Accelerated approval explicitly permits earlier decisions using surrogate endpoints (U.S. Food and Drug Administration 2024; Gyawali, Hey, and Kesselheim 2019).	Survival, function and quality of life may require years. In one 2009–2013 accelerated-approval cohort, half of required postapproval studies were completed only after at least 3 years [M] (Naci, Smalley, and Kesselheim 2017); later oncology analyses find unresolved clinical benefit beyond 5 years in some cases [M] (Liu, Kesselheim, and Cliff 2024).	Disease progression and sufficiently powered clinical follow-up.	Potentially very large, but the number of consequential decisions per target horizon is program-specific [NQ].

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Table 1 continued

Domain	**Fast loop / proxy**	**Current AI cycle**	**Human / classical cycle**	**Target-level evidence τ_{T}**	**Limiting cycle**	**Approx. N_{proxy} / status**
Reward-model alignment	A learned reward model, preference model, or direct preference loss is queried at every optimization step.	Optimization feedback is available every policy update; the useful independent axis is optimization budget or KL distance rather than wall-clock time [M/NQ] (Gao, Schulman, and Hilton 2022; Rafailov et al. 2024).	Human judgments are sufficiently costly that they are commonly distilled into a learned evaluator; no universal annotation latency exists [NQ].	Fresh human judgments or an independently held "gold" evaluator.	Human/evaluator throughput and coverage of the shifted policy distribution.	Directly variable experimentally via optimization strength.
Cybersecurity	Sandbox exploit success, tests, CTF flags, crash/reproduction signals.	CVE-Bench provides reproducible sandboxed real-world CVEs; the best reported agent framework exploited up to 13% of benchmark vulnerabilities [M] (Zhu et al. 2025).	Human security researchers also obtain fast reproduction feedback once a concrete vulnerability is isolated, but discovery and incident response times vary widely [NQ].	Real-world exploitability, prevalence, attacker adaptation and absence of undiscovered attack paths are not fully certified by sandbox success.	For a known exploit: execution time. For "system is secure": no finite exhaustive cycle in general.	[NQ]
Recommendation / engagement optimization	Clicks, likes, watch time and other engagement responses are available almost immediately.	Recommendation systems can optimize real-time engagement [M] (Khambatta et al. 2023).	Human editors, advertisers and media organizations have analogous short-term audience metrics, but no matched cycle estimate is needed here [NQ].	User benefit or "time well spent" need not agree with engagement. In a preregistered experiment, actual-preference recommendations obtained more clicks (52% vs. 40%), while ideal-preference recommendations improved several measures of perceived user benefit [M] (Khambatta et al. 2023).	Depends on the chosen target; some welfare effects can be measured immediately, others only longitudinally.	Not necessarily large.
Public benchmarks and leaderboards	Benchmark score is available immediately after evaluation.	Models and prompts can be repeatedly selected against public or semi-public benchmark feedback.	Exams, scientific metrics and industrial KPIs provide analogous rapidly scored objectives.	External validity, contamination resistance and transfer to deployment may require held-out evaluation or later real-world evidence.	Independent held-out tasks or deployment.	Potentially large but usually not observable from public data.

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Table 1 continued

Domain	Fast loop / proxy	Current AI cycle	Human / classical cycle	Target-level evidence τ_T	Limiting cycle	Approx. N_{proxy} / status
Macroeconomic policy	Financial conditions, market rates, expectations and monthly/quarterly indicators provide relatively fast intermediate signals.	AI-assisted analysis can be generated much faster than macroeconomic effects unfold, but no defensible generic AI policy-update latency exists [NQ].	Central banks update policy and forecasts repeatedly while awaiting full transmission.	Federal Reserve summaries of the literature place maximal effects of a monetary-policy shock on activity and inflation at roughly 1–2 years [M] (Kugler 2025); the ECB likewise describes long, variable and uncertain lags (European Central Bank 2024).	Adjustment of contracts, borrowing, investment, wages, prices and expectations in the real economy.	[NQ]
Rare-event safety	Near misses, simulations, checklists, component tests and probabilistic risk models provide fast proxy evidence.	AI can run simulations and automated safety checks rapidly, but rare real failures remain event-limited [S/NQ].	A general-aviation accident investigation often requires 12–24 months to complete [M] (National Transportation Safety Board 2024); nuclear regulation explicitly supplements sparse operating experience with probabilistic risk assessment and precursor analysis (U.S. Nuclear Regulatory Commission 2024a; U.S. Nuclear Regulatory Commission 2024b).	The strongest evidence may require occurrence of a rare accident, followed by causal investigation.	Event frequency plus investigation latency.	Potentially enormous or undefined.
AI alignment and long-horizon safety	Behavioral evals, reward models, red-team suites, automated graders and held-out benchmarks.	Individual evals can be cheap and repeated, but no empirical literature currently supports a general target-evidence latency for robustness to future capability growth, strategic adaptation or deployment shift [NQ].	The closest human analogues are safety assurance, auditing, probabilistic risk assessment and incident investigation.	The relevant target may only be tested under new capabilities, distribution shift, prolonged deployment or strategic pressure.	Unknown; partly event- and capability-transition-limited.	**Not yet empirically quantified.**

5.1 Reading the Matched Horizons

Table 1 is easier to read once the domains are grouped by *why* their feedback geometry differs, not only by field name.

Matched verifiers. When the fast loop already checks the target criterion, a very large search or update count is compatible with safety. Formal theorem proving and rule-complete games are the clearest cases: Lean acceptance, terminal win/loss, and similar verifiers make proxy and target coincide, so effective $N_{\mathrm{proxy}} \simeq 1$ even when internal search depth is enormous. The separate risk is semantic: a perfect checker on a mis-formalized statement is still the wrong target.

Mixed fidelity inside one artifact. Executable software often combines both regimes in a single codebase: $\tau_{\mathrm{P}} \approx \tau_{\mathrm{T}}$ for well-covered functional tests, but $\tau_{\mathrm{P}} \ll \tau_{\mathrm{T}}$ for security, maintainability, performance under real load, and rare bugs. Low-level algorithm optimization and chip floorplanning show a related *proxy hierarchy*: dense feedback from correctness checks, learned value estimates, or placement surrogates, while the stronger target—measured hardware latency, timing closure, silicon performance—is sampled only sparsely or arrives only downstream. AI can shorten construction time without making every downstream property available in the fast reward.

Clocks set by the world. Weather forecasting and general forecasting illustrate a different bottleneck. AI can compress candidate generation dramatically—GraphCast produces a ten-day forecast in under a minute—but prospective target evidence still waits on atmospheric evolution or event resolution. Neither humans nor models can manufacture the answer early; better reasoning can improve forecasts without closing the prospective target loop. This is where N_{proxy} can grow mechanically as optimization improves while τ_{T} stays fixed.

Shrinking τ_{T} through automation. Autonomous materials synthesis and closed-loop chemistry show the opposite response to physical delay. Self-driving laboratories do not merely generate hypotheses faster; they move characterization into the loop through robotics, in-situ measurement, and automated next-step selection. The limiting cycle becomes handling, reaction kinetics, and measurement rather than human handoff. Progress here attacks the target-evidence latency itself, not only τ_{G} or τ_{P}.

Institutional fast-proxy / slow-target structures. Clinical medicine is the strongest real-world example: surrogate endpoints can guide decisions years before survival, function, or quality of life are known, and regulators explicitly treat surrogates as predictors rather than direct measures of benefit. Macroeconomic policy provides a human analogue of acting repeatedly under delayed target effects—central banks revise policy on monthly or quarterly signals while full transmission to activity and inflation may take one to two years—but confounding shocks and a moving environment prevent a clean causal read on N_{proxy}.

Where accumulation under proxy selection shows up directly. Reward-model alignment is the strongest direct AI evidence for accumulation under proxy selection: proxy scores can keep rising after independently evaluated quality has peaked and begun to decline. Public benchmarks add a different lesson: evaluator *independence* matters. A fast public score can lose informativeness once optimization is conditioned on its blind spots; dynamic or held-out evaluation is partly a response to that failure mode. Recommendation and engagement optimization are an important counterexample to a purely temporal theory: proxy failure can occur even when N_{proxy} is not large, because engagement and user benefit diverge immediately.

Local verifiers and open-ended targets. Cybersecurity combines unusually strong local feedback—sandbox reproduction, crash signals, CTF success—with a global target that has no finite exhaustive cycle in general. Demonstrating one exploit is much easier than certifying absence of exploitable paths. Rare-event safety is the limiting case of the horizon problem: because direct target observations are rare, mature fields construct surrogate feedback systems—probabilistic risk assessment, precursor analysis, investigation protocols—rather than waiting for repeated catastrophes.

Alignment as application, not evidence. The alignment row is the motivating application of the paper, not independent evidence for it. Reward-model overoptimization demonstrates the local mechanism; rare-event safety and clinical surrogate endpoints provide institutional analogues. Any stronger alignment extrapolation should remain explicitly conditional until target-evidence latency under capability growth, strategic adaptation, or deployment shift can be measured rather than assumed.

Two acceleration patterns. Two patterns recur across the table. First, AI can shrink both generation and target verification when the target is computationally checkable—formal proof, covered software properties, and terminal game outcomes. Second, AI can shrink generation or proxy evaluation while leaving target evidence nearly unchanged—weather, forecasting, clinical outcomes, and many deployment properties. The second regime is where faster optimization raises exposure unless independent target checks keep pace.

6 Feedback Timing in Human Learning

Verifier construction can look like a recommendation to shorten every loop. Human learning results caution against that reading.

Educational feedback meta-analyses find medium average effects with large heterogeneity; information content strongly moderates benefit, and some feedback forms outperform others (Wisniewski, Zierer, and Hattie 2020). Meta-analyses of knowledge-of-results timing in motor learning find statistically significant timing effects with comparatively small effect sizes (Travlos and Pratt 1995). Delayed judgments can improve metacognitive accuracy by making long-term memory more diagnostic (Rhodes and Tauber 2011). Some test-learning experiments find delayed feedback beating immediate feedback (Butler, Karpicke, and Roediger 2007).

τ_{P} alone does not predict learning rate. A fast signal with little target information can underperform a slower, more diagnostic one. The human analogue is not "immediate feedback is good." Learning tracks the information carried by feedback, and timing sets how much misleading optimization can run before better information arrives:

$$\text{learning depends on the information carried by feedback,}$$

with timing entering through the exposure window before correction.

The same caution applies to constructed verifiers. Earlier observation helps when the earlier signal is informative about the target. It can hurt when it is a more exploitable surrogate.

7 The AI–human Analogy Is Structurally Asymmetric

The matched examples in this paper should not be read as claiming that AI optimization is simply accelerated human problem solving. The comparison is useful because the same feedback structures recur in both cases, but several important mechanisms are asymmetric.

AI has much greater parallel search. A human expert normally explores a small number of candidate solutions at once. Digital systems can generate, execute, and reject enormous candidate sets in parallel. AlphaProof trains on millions of formally checked problems (Hubert et al. 2025); AlphaDev explored millions of candidate programs (Mankowitz, Michi, Zitnick, et al. 2023). Consequently, wall-clock ratios such as $\tau_{\mathrm{T}}/\tau_{\mathrm{P}}$ can substantially understate the amount of selection occurring between target checks. This is one reason we define N_{proxy} primarily as the number of consequential proxy-guided updates, rather than as a time ratio.

AI state can be copied and branched cheaply. Human learning is transmitted imperfectly through teaching, writing, imitation, and institutions. A trained digital state can instead be copied, checkpointed, branched, evaluated, and recombined with comparatively little loss. A useful update can therefore propagate across many simultaneous search processes without requiring each process to rediscover it. There is no close human analogue at the level of individual cognition.

AI can be optimized internally. Human feedback primarily selects actions, strategies, institutions, or people. Machine learning also changes the internal parameters that generate subsequent behavior. Gradient-based optimization can therefore perform many small internal updates between externally meaningful decisions. Whether each gradient step should count toward N_{proxy} depends on the analysis. For safety questions, the relevant unit is normally the smallest update capable of producing a consequential change in the system's policy or deployment behavior, not every floating-point parameter update.

Human progress has stronger examples of verifier invention. The asymmetry also runs in the other direction. Human scientific progress repeatedly changed not only candidate solutions but the *measurement regime*: new instruments, standardized measurements, controlled experiments, statistical methods, and institutions made previously weakly observable claims testable. Current AI systems can propose experiments and help design measurement procedures, and self-driving laboratories automate existing instances (Volk et al. 2023; Szymanski et al. 2023). But the historical record contains much stronger examples of humans changing what could be observed at all.

Humans can sometimes learn from $N = 1$ events. Wars, accidents, financial crises, constitutional failures, and scientific anomalies may occur too rarely for ordinary repeated optimization. Humans nevertheless build causal narratives, compare mechanisms across cases, construct counterfactuals, and change institutions after unique events. These updates can be wrong, but they are not well represented as repeated trials from a stationary distribution. Present AI training, by contrast, derives much of its strength from large repeated datasets or synthetically generated task families.

Human institutions can deliberately separate evaluators. Science, auditing, courts, safety engineering, and other institutions often create partially independent channels of criticism. Their purpose is not merely to increase feedback frequency but to prevent one selection criterion from controlling all evidence about itself. AI evaluation often compresses many judgments into a single benchmark or reward model. Multiple AI critics do not automatically recreate institutional independence if they share training data, architecture, prompts, or failure modes.

The comparison is therefore conditional. The paper makes a comparison at the level of feedback architecture:

$$\text{candidate} \to \text{selection signal} \to \text{update} \to \text{later target evidence.}$$

It does not claim equivalence between human cognition, scientific institutions, and machine learning. The important empirical question is narrower: holding the target and evaluator structure approximately fixed, what changes when the number of proxy-guided selections between independent target checks increases?

This scope restriction matters for interpreting the historical examples. The human record motivates the variables and provides useful analogies, but it does not by itself establish a causal law for AI systems. Conversely, reward-model overoptimization provides direct evidence for the accumulation mechanism in AI but does not establish that institutional or scientific systems obey the same quantitative relationship.

8 Discussion

The companion paper's operative distinction is whether optimization runs inside a loop whose evaluator is both fast and tied to the target (Zarncke 2026). This paper's operative distinction is whether that loop can be rebuilt.

Capability improvements search the solution space faster. Measurement, instrumentation, testing, and experimental design make the consequences of that search legible faster. The two are complements. Fast search with a strong verifier produces the favorable regime of theorem proving and well-tested software. Fast search with a weak but rapidly computed proxy produces the regime of reward-model overoptimization.

The table does not establish a causal law across domains. Reward-model overoptimization is direct evidence for accumulation under proxy selection. Clinical surrogates, rare-event safety, and self-driving laboratories are structural analogues and existence proofs for construction. The alignment row is the motivating application, not independent evidence.

9 Conclusion

Verifier construction is the attempt to move target-relevant error into the loop that actually selects updates. It can proceed by earlier observation, cheaper repetition, or decomposition. Replacing a slow target with a fast surrogate is construction too, and it is the move that creates the horizon gap.

Which of those moves is available depends on the feedback regime. Exact symbolic and well-covered executable feedback already place the local target in the fast loop. Repeated empirical feedback can be accelerated by instrumentation, but usually not to zero. Noisy statistical, sparse, and endogenous regimes are where construction is both most needed and most likely to produce a misleading surrogate.

The test of an evaluation architecture is not whether a score exists. It is whether target-relevant errors become visible at roughly the rate at which optimization can discover and amplify them.

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